REALIZATION CALCULUS

Komorebi Physics

Continuum, Geometry and Refinement

FIELDS / STRUCTURE / LIMITS

Komorebi P.

2026

KOMOREBI PHYSICS

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Continuum, Geometry and Refinement

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A mathematical account of resolved dynamics, free fields, gauge transport, gravity, arithmetic structure, and cosmological models.

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ABOUT THIS VOLUME

Mathematics, models, and evidence

The volume brings together a shared mathematical language and several physical applications. Projector identities and transfer maps lead to complete state spaces, unbounded operators, compatible refinements, and controlled comparisons between descriptions.

Written proofs, recorded compiler checks, finite computations, and physical interpretations have different roles. The mathematical chapters state their hypotheses. The source studies identify the formal declarations, recorded evidence, and remaining verification limits. The visual studies illustrate specified examples rather than replace their proofs.

The free Dirac group, maximal generator domain, and strong Schrödinger equation have compiler-checked proofs. The centered Fourier family also has checked graph-domain and strong-resolvent limits and quadratic bounds under a finite third-order moment. The abstract spectral-functional-calculus identification remains a written deduction without a separate Lean API linkage. Physical-model chapters retain the production-energy obstruction and their stated assumptions.

The equation, theorem, figure, and local section numbers belong to their named treatment. The contents and book folios provide a single navigation system. Source identifiers are retained in the verification material where they are needed for reproducibility.

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INTRODUCTION

A common language for resolved physics

A physical description often begins by choosing what to resolve. A state may be separated into spatial cells, Fourier modes, spin channels, or arithmetic registers. The useful questions begin after that choice: how is the state reconstructed, how do its components exchange information, and what is retained when the resolution changes?

This book develops those questions through a sequence of concrete constructions. Projector and transfer identities supply the finite algebra. Hilbert-space completion and operator domains supply the analytic control. The free Dirac model carries that route through to a self-adjoint Hamiltonian and the identification of its unitary evolution.

Gauge transport and constrained gravity introduce further requirements. Paths must keep their order, constraints must retain their hypotheses, and a comparison between resolutions must name its topology. Arithmetic and polyphase refinement show how compatible embeddings differ from elimination of hidden degrees of freedom. The latter can create self-energy and memory rather than a smaller autonomous copy of the original dynamics.

The final physical-model chapters ask what follows when a refinement fiber is coupled to spacetime and used in cosmology. Their interpretation is controlled by the production-energy test. A consistent transport calculation from a prescribed distribution does not establish that a microscopic crossing produced that distribution.

The chapters share a language, not a claim that all their physical problems have the same solution. Exact identities, analytic limits, compiler certificates, numerical examples, and proposed interpretations remain distinguishable. The book can therefore be read as one mathematical argument without treating every application as an established theory of nature.

READING THE BOOK

The argument and its conventions

From local structure to complete spaces

Read the first part in order. Packet identities establish what is conserved. Fourier completion makes the state space complete. The operator-domain argument makes the Hamiltonian meaningful. The generator proof then connects that operator to the constructed evolution.

From structure to a physical model

The middle parts change the model while retaining the same discipline: name the state space, the transfer or refinement map, the mathematical assumptions, and the residual obstruction. In the matter and cosmology part, the distribution used for transport remains an explicit input until a controlled production calculation supplies it.

Notation and references

Each named treatment retains its local section, equation, figure, and theorem numbering. References to those numbers are local to that treatment unless another title is named. Preliminary pages use Roman numerals. Arabic page numbers run continuously through the main text and technical appendices. The PDF bookmarks provide detailed navigation within each treatment. The running headers link back to the contents.

Proof and verification

The mathematical chapters carry their hypotheses and written proofs. The visual studies illustrate declared examples. The technical appendices identify the formal statements and recorded computations, including missing evidence and uncompiled extensions. A document build is not a theorem check.

The free-field comparison

The continuum and centered Fourier families use the same infinite mode index set. A changing finite sampled lattice is not silently substituted for that setting. The first-order Hamiltonian domain and the third-order moment required by the quadratic rate are also kept distinct.

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